Original Investigation

Evaluating the Efficacy of Perplexity Scores in Distinguishing AI-Generated and Human-Written Abstracts

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Rationale and Objectives: We aimed to evaluate the efficacy of perplexity scores in distinguishing between human-written and AI-generated radiology abstracts and to assess the relative performance of available AI detection tools in detecting AI-generated content.

Methods: Academic articles were curated from PubMed using the keywords "neuroimaging" and "angiography." Filters included English-language, open-access articles with abstracts without subheadings, published before 2021, and within Chatbot processing word limits. The first 50 qualifying articles were selected, and their full texts were used to create AI-generated abstracts. Perplexity scores, which estimate sentence predictability, were calculated for both AI-generated and human-written abstracts. The performance of three AI tools in discriminating human-written from AI-generated abstracts was assessed.

Results: The selected 50 articles consist of 22 review articles (44%), 12 case or technical reports (24%), 15 research articles (30%), and one editorial (2%). The perplexity scores for human-written abstracts (median; 35.9 IQR; 25.11–51.8) were higher than those for AI-generated abstracts (median; 21.2 IQR; 16.87–28.38), ($p = 0.057$) with an AUC = 0.7794. One AI tool performed less than chance in identifying human-written from AI-generated abstracts with an accuracy of 36% ($p > 0.05$) while another tool yielded an accuracy of 95% with an AUC = 0.8688.

Conclusion: This study underscores the potential of perplexity scores in detecting AI-generated and potentially fraudulent abstracts. However, more research is needed to further explore these findings and their implications for the use of AI in academic writing. Future studies could also investigate other metrics or methods for distinguishing between human-written and AI-generated texts.

Key Words: Artificial intelligence; Natural Language Processing; Large language model; Perplexity score; GPT.

Abbreviations: LLM Large Language Model, **GPT** Generative Pre-trained Transformer, **PPL** Perplexity scores, **ROC** Receiver Operating Characteristic, **AUC** Area under the ROC curve

INTRODUCTION

Large Language Models (LLMs) have recently garnered substantial attention across multiple fields (1–5). Trained on extensive and diverse datasets, these models are increasingly adept at producing language that closely mirrors human communication.

The use of LLMs in generating scholarly manuscripts has surged, with an estimated 1% of all scholarly articles (approximately 60,000 papers) in 2023 being LLM-assisted (6). While exact figures remain elusive, there has been a noticeable increase in the use of LLMs by "paper mills" to produce substandard or deceptive articles. Furthermore, fraudulent papers are also being generated with fake citations (7,8). In response, many publishers are developing guidelines to ensure the ethical use of LLMs in research (6,9,10).

Despite advancements in plagiarism detection technologies, the effectiveness of current methods in identifying LLM-generated manuscripts is still under question. Web-based AI detection tools have been developed to identify LLM-generated manuscripts, yet their reliability remains uncertain (11,12). Perplexity scores, a metric originally

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introduced to assess the predictability and complexity of text, have been employed in various studies to evaluate the likelihood of text being generated by an LLM as opposed to a human (13–16). However, the application of perplexity scores to academic abstracts, particularly within the field of radiology, has not been thoroughly investigated.

Our primary aim was to evaluate the efficacy of perplexity scores in distinguishing between human-written radiology manuscript abstracts available in the literature and AI-generated abstracts that we created using the associated published manuscripts. We secondarily aimed to assess the relative performance of available AI detection tools in detecting AI-generated content.

MATERIAL AND METHODS

This study was conducted with publicly available open-access articles and did not require institutional IRB approval.

Selection of Articles

We used the PubMed search engine, with a specific focus on the MEDLINE database, to identify manuscripts through a search with the keywords "***neuroimaging***" and "***angiography***."

To ensure the relevance of the results and maintain project feasibility, several filters were applied. A word limit of 4000 was set to align with the constraints of the ChatGPT 3.5 prompts, and the search focused on English-language publications. The search was further restricted to open-access articles published before 2021. Because the ChatGPT training dataset does not include data after September 2021 and generative AI models were not yet publically available during this period. Therefore, the selected articles were presumably exclusively authored by humans.

Abstracts with subheadings were excluded due to the variability in this format across journals, which could complicate our creation of AI-generated abstracts, and given the challenge of maintaining consistent word counts for each subheading.

The search results were sorted using PubMed's "Best Match" option. The final selection of papers was conducted independently by two authors. The articles were converted from PDF to Word format to facilitate copying, pasting, and word count calculation. The full texts and abstracts of the 50 selected articles were saved for further analysis.

The text of the saved articles was extracted without making any sentence changes or altering the writing format. Received/acceptance dates, author names, author contact information, acknowledgments, references, funding, and conflicts of interest were not included. Visual elements, including figures and tables, that could hinder copy-paste operations were not included, while their captions were included.

Generation of AI-written Abstracts

The ChatGPT3.5 model (17) was used to generate abstracts for selected articles. The original abstracts, referred to as *"human-written abstracts",* were removed from the articles, and the remaining sections (introduction, methods, results, discussion, conclusion) were copied and pasted into the ChatGPT3.5. A fixed prompt was used for each article text. The prompt was: ***"Please write an abstract for my article titled '…' There should be no subheadings in the abstract you write, and it should contain … words. Here's my article: …"*** This prompt, written in English, was used to instruct ChatGPT3.5 to generate an abstract with the same word count as the original abstract for the submitted article (Supplementary Fig 1). No additional information was provided in the text transferred to ChatGPT3.5 other than the prompt, and no further details were given about the articles. Additional prompts were not used to modify or improve the model's response. The history was reset after each task to avoid bias and information leakage. The ChatGPT 3.5 web access feature that allows it to connect to the internet was turned off and no ChatGPT plugins were active.

Abstracts Perplexity Score

Perplexity is a metric used to quantify a language model's ability to predict text, serving as an indicator of the text's originality. AI-generated abstracts are generally more predictable and thus yield lower perplexity scores, while human-written abstracts, with their higher originality, tend to result in higher perplexity scores (13). To generate the perplexity score, each abstract was passed through the pre-trained Llama 2 large language model from the Hugging Face library (18), a popular open-access platform dedicated to developing and maintaining Transformers for various natural language processing tasks. It's important to note that the chosen language model significantly influences the perplexity score, as models differ in terms of training data, architecture, and size. We chose the Llama 2 model due to its known high level performance (18) and its open-access availability. We ran the Llama 2 model on Google Colaboratory, which was chosen for its capabilities in terms of RAM and disk space.

The perplexity for each sentence was calculated. We then averaged these scores to obtain a single perplexity score for the entire abstract. The mathematical formula for perplexity is as follows:

$$\mathrm{PPL}(X) = \exp\left\{-\frac{1}{t}\sum_{i}^{t}\log p_\theta\left(x_i \mid x_{<i}\right)\right\}$$

Where θ is the LLM (Llama-2) and $\log p\theta(xi|x < i)$ is the log-likelihood of the i^{th} token prediction, based on the preceding tokens (13,19,20).

In essence, lower perplexity scores indicate that the language model finds the text to be more predictable or coherent based on its training data, suggesting that the text aligns well with the model's learned language patterns. For example, if a language model assigns a high probability to the correct next word in a sequence, the perplexity will be lower, implying that the model is more "confident" in its predictions. Conversely, a higher perplexity score indicates that the model is more uncertain, suggesting that the text might be less coherent or more complex.

To further assess the effectiveness of perplexity scores in distinguishing between human-written and AI-generated abstracts, we used the *Receiver Operating Characteristic (ROC)* to identify the optimal perplexity score threshold for differentiation.

Evaluation with AI Detection Tools

To evaluate the ability of LLMs to distinguish between AI-generated and human-written abstracts, we presented both types of abstracts (human-written and AI-generated) to Google's Bard, which is now known as Gemini as of February 8, 2024. We chose Bard's model because Microsoft's Bing model, which is now known as Copilot as of July 2023, tended not to respond during the test, and the ChatGPT model is already being used to generate abstracts. The prompt entered into Google's Bard is shown in Supplementary Figure 2. No additional information or guidance beyond the articles and prompts entered into the chat was provided to Google Bard. The selections made by Google Bard during the test and the rationale behind those selections were documented. Responses provided by Bard to human-written and AI-generated abstracts, along with their justifications, were also recorded. An example of how Bard evaluated its selections is shown in Table 1.

We chose the ZeroGPT (12) and GPTZero (11) AI detection tools for our experiment. These tools were selected because they are freely available and can provide a clear binary result (0 for human-written, 1 for AI-generated) as well as a probability indicating the percentage of AI-generated content. We applied these tools to a set of 100 abstracts (50 human-written and 50 AI-generated), noting both the final binary results and the probabilities for each abstract.

In evaluating the performance of these tools, we considered five different metrics: specificity, sensitivity, accuracy, balanced accuracy, and ROC AUC score. We also calculated the 95% confidence intervals (CIs) for accuracy, sensitivity, specificity, and balanced accuracy. The formulas for these metrics are as follows: Sensitivity = TP/(TP+FN), Specificity = TN/(TN+FP), Balanced Accuracy = (Sensitivity + Specificity)/2 (21).

The terms TN, TP, FP, and FN represent true negatives, true positives, false positives, and false negatives, respectively. The results for sensitivity, specificity, accuracy, and balanced accuracy are determined based on the final binary outcome provided by the AI detection tools. On the other hand, the ROC and AUC values are constructed based on the probability outcomes provided by the model.

Statistical Analysis

The Chi-square ($\chi 2$) test was used to compare categorical variables. The perplexity scores of human-written and AI-generated abstracts were compared using the Mann–Whitney U test. The performance of AI detection tools, as indicated by AUC values, was compared using the DeLong test, which compares the differences between paired AUC values and their standard errors to calculate a p-value.

A p-value of ≤0.05 was considered statistically significant. All analyses were conducted using either Microsoft Excel (version 14.7.1, Microsoft Corp, Redmond, WA, USA), SPSS software (version 28; IBM Corp., Armonk, NY, USA), or R software (version 4.1.2; The R Foundation for Statistical Computing, Vienna, Austria).

RESULTS

To identify 50 eligible articles, we iterated through our selection process, which required a manual review of 260 abstracts (**Fig 1**). The selected 50 articles consist of 22 review articles (44%), 12 case or technical reports (24%), 15 research articles (30%), and one editorial (2%) (**Supplementary Table 1**).

Human-Written vs. AI-Generated Radiology Abstract Perplexity Scores

The perplexity scores as calculated using Llama 2 for human-written abstracts (median; 35.9 IQR; 25.11–51.8) were higher than those for our AI-generated abstracts (median; 21.2 IQR; 16.87–28.38), with a p-value of 0.057. Based on the perplexity scores of the abstracts, the best threshold for determining the source of an abstract is found to be 22.04015. The detection performance of the perplexity score was found to be AUC = 0.7794 (**Fig 2**).

AI Detection Tool Efficacy

In our experiment to find AI-generated abstracts using Google Bard, Bard performed less than chance in identifying, 36% of the AI abstracts ($p > 0.05$). Google Bard reported that it made its selections by examining various aspects (writing style, fluency, grammar, grammatical accuracy, emphasis, emotional expression, creativity, and originality).

TABLE 1. This Table Shows the Results of Examining AI Detection Tools Using Different Evaluation Metrics

Model	Accuracy* (%)	Sensitivity* (%)	Specificity * (%)	Balanced accuracy* (%)	ROC AUC score
ZeroGPT	0.69 (0.59–0.77)	0.82 (0.67–0.90)	0.56 (0.41–0.70)	0.69 (0.59–0.76)	0.586
GPTZero	0.95 (0.87–0.98)	1	0.9 (0.78–0.96)	0.95 (0.89–0.97)	0.868

*Values in parentheses indicate a 95% confidence interval.

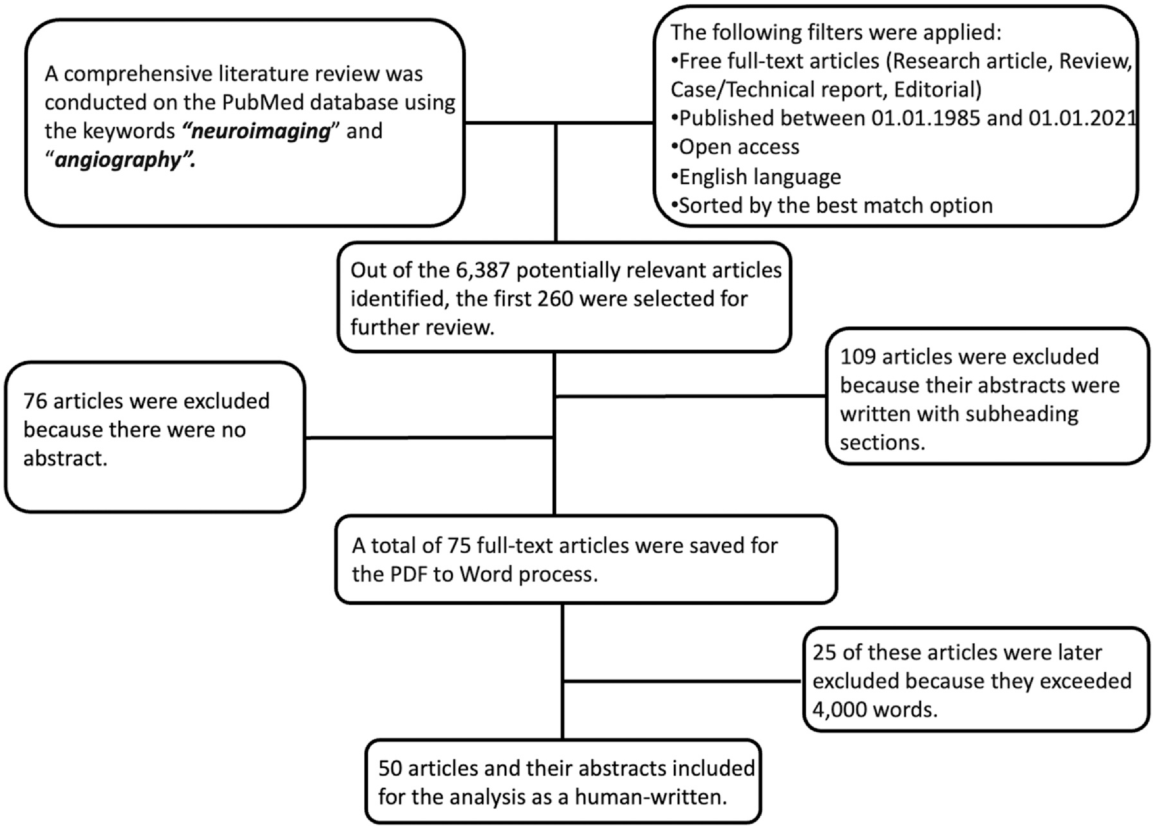

Figure 1. Article selection procedure of the study for the source of abstract detection.

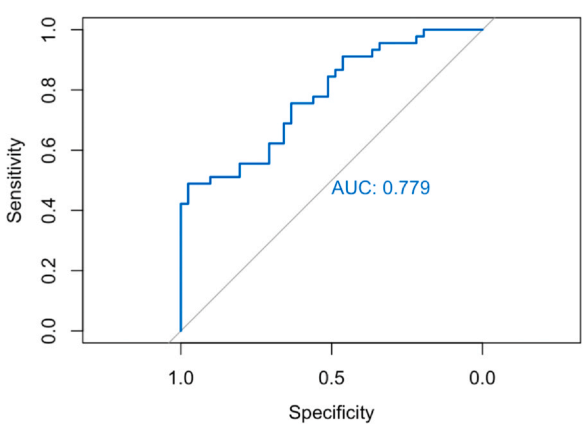

Figure 2. This figure visualizes the ROC and AUC values of our experiment, where we attempt to determine whether abstracts are human-written or AI-generated based on their perplexity scores.

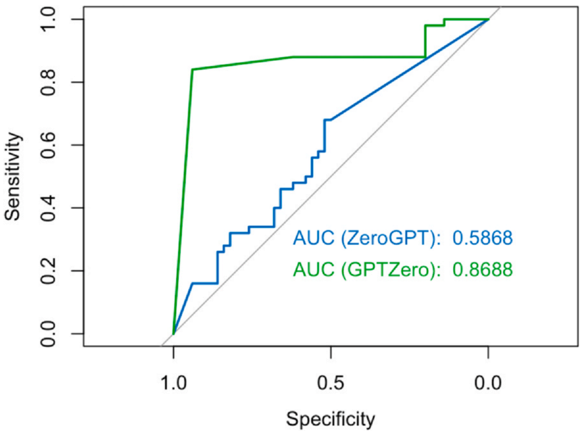

Figure 3. The performance of ZeroGPT and GPTZero in determining whether abstracts are human-written or AI-generated is illustrated through the ROC and AUC values.

The reasons for Bard's correct and incorrect predictions examples are shown in **(**Supplementary Table 2**)**. The detection performance of ZeroGPT and GPTZero is summarized in Table 1. It is known that the GPTZero model, which also uses perplexity scoring in detection technology, has an AUC value of 0.8688 for detection performance. This is statistically significantly higher than the AUC value of ZeroGPT, which is 0.5868 ($p < 0.001$), as shown in Figure 3**.**

DISCUSSION

In this study, we found that the perplexity scores of human-written abstracts were higher than those generated by AI, as expected, which suggests that AI-generated abstracts and potentially fraudulent manuscripts may be frequently identifiable upon submission. Furthermore, our findings suggest that LLMs are not directly capable of making this distinction, with one studied tool identifying only 36% of AI-generated abstracts correctly, but more complex tools that use multiple methods including perplexity score calculation can have an accuracy of up to 95% and AUC of 0.87.

Recently, some published articles were found to contain AI-generated sentences that were not identified by editors. Even though these sentences were identified and retracted later (22), they may harm the academic world. This highlights the need to develop effective and efficient tools for detecting AI-generated content. Determining human-written from AI-generated text may be difficult for human reviewers. In a prior experiment also involving 50 abstracts, human reviewers correctly identified only 68% of the AI-generated abstracts and 86% of the human-written abstracts (23). Similarly, in a study conducted by Lawrence et al., which involved 20 abstracts, six blinded orthopedic surgeons were unable to distinguish whether abstracts were human-written or AI-generated ($p = 0.999$) (24).

Given that it's currently challenging for human reviewers to make this distinction, various solution methods have been attempted. Rashidi et al. evaluated 1000 human-written abstracts from well-known journals using a RoBERTa-based detector from Open Ai's Hugging Face repository. Their study revealed that the model incorrectly classified 8% of the human-written abstracts as AI-generated (25). In recent days, commercial online tools have emerged. However, their effectiveness remains uncertain due to a lack of comprehensive studies. Bellini et al. (26) conducted a comparison of four such online tools: GPTZero, ZeroGPT, Writer ACD, and Originality. They found that these tools varied in performance and cautioned about the potential for these tools to generate false positives or overlook true negatives (26). However, unlike our study, these studies did not consider the detection probabilities generated by these online tools. In our evaluation, GPTZero, which also used perplexity scoring for detection, exhibited better performance than ZeroGPT. Furthermore, the fact that these tools are not open-source, and their internal mechanisms are not fully understood creates a “black box” scenario. This lack of transparency hinders the complete resolution of this issue.

Our study primarily focuses on the concept of perplexity score. Perplexity scores fundamentally determine how well a probability model predicts a sample. In other words, perplexity scores quantify how well the model understands the text it's generating. A lower perplexity score indicates that the model is more certain about its predictions. Keukeleire et al., in their study analyzing TV show scripts, found that the most realistic and coherent AI-generated texts had lower perplexity values (13).

Our findings support the idea that perplexity can be a valuable metric in these contexts. Although we cannot definitively determine the reason, it is important to note that GPTZero's strong performance might be correlated with its APIs or training data being aligned with OpenAI solutions. Additionally, while perplexity calculation could potentially serve as a detection feature, its implementation into programs could be challenging. To establish universal distinguishing thresholds or cut-off values for perplexity scores-based detection, experiments should extend beyond the field of radiology and include a larger set of abstracts.

This study has several limitations. First, we did not utilize prompt engineering (such as zero-shot prompting, few-shot prompting, instruction following, and chain-of-thought) (27–30) when generating abstracts from original articles, so our AI-generated abstracts may have had lower perplexity scores than what is achievable with more advanced prompts. Additionally, while developing AI-generated abstracts, we faced limitations due to word count restrictions and the inability to upload tables/figures. Future improvements in these areas by Chatbots could lead to more generalizable results. We also utilized only abstracts without subheadings, likely skewing our selection toward review articles and away from scientific articles published in the major radiology journals, so generalizability may be limited. Although the Llama 2 model we used for perplexity calculation is relatively strong among open-access models, comparative studies will be necessary in this competitive and rapidly evolving field.

CONCLUSION

In conclusion, this study underscores the potential of perplexity scores in detecting AI-generated and potentially fraudulent abstracts. However, more research is needed to further explore these findings and their implications for the use of AI in academic writing. Future studies could also investigate other metrics or methods for distinguishing between human-written and AI-generated texts.

DECLARATION OF COMPETING INTEREST

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

APPENDIX A. SUPPORTING INFORMATION

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.acra.2025.01.017.

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